



EWM226-900H30S Product Specification Sheet

AT commands 868/915MHz LoRa Wireless Module



Table of contents

Disclaimer and Copyright Notice	1
Chapter 1 Product Overview	2
1.1 Product Introduction	2
1.2 Features and Functions	2
1.3 Application Scenarios	2
Chapter 2 Specifications	3
2.1 Product Selection Comparison	3
2.2 Basic parameters	3
Chapter 3 Mechanical Dimensions and Pin Definitions	4
3.1 Mechanical Dimensions and Pin Definitions	4
Chapter 4 Recommended Connection Diagram	6
Chapter 5 Functional Details	7
5.1 Working Mode	7
5.1.1 Transmission mode	7
5.2 Serial port firmware upgrade	11
5.2.1 Upgrade your host computer quickly	11
Chapter 6 AT Commands	13
6.1 General instructions	13
6.1.1 AT+RESET soft reboot	13
6.1.2 AT+DEFAULT to restore factory settings	13
6.1.3 Use AT+DEVTYPE to query product model	13
6.1.4 AT+FWCODE Query Software Version	13
6.1.5 AT+ IAP Serial port firmware upgrade	14
6.1.6 AT+MOD Transmission Working Mode	14
6.1.7 AT+ADDR Communication Address (Match Filtering)	14
6.1.8 AT+NETID Communication Network Code (Matching Filtering)	14
6.1.9 AT+CHANNEL Communication Channel (Carrier Frequency)	15
6.1.10 AT+KEY Communication Password (Data Encryption)	15
6.1.11 AT+UART Serial Port Baud Rate and Verification	15
6.1.12 AT+PACKET Serial Port Packing Length	15
6.1.13 AT+URXT Serial Port Frame Drop Conditions	16
6.1.14 AT+RATE Wireless Transmit/Receive Rate	16
6.1.15 AT+UFREQ Setting the operating frequency parameters	16
6.1.16 AT+LBT Listen before you speak (avoidance)	17
6.1.17 AT+LBR Listen-before-send Judgment Condition	17
6.1.18 AT+ROLE HOP characters	17
6.1.19 AT+WTIME Slave Data Transmission Cycle	17
6.1.20 AT+ERSSI Ambient Signal Strength	18
6.1.21 AT+DRSSI Received Data Packet Signal Strength	18
6.1.22 AT+HPASW High Power Switch	18
6.1.23 AT+POWER power rating	18
6.1.24 AT+FLIST HOP Mode Frequency List	19
6.1.25 AT+SYNCT HOP slave synchronization timeout	19

6.2 Signaling testing (certification)	20
6.2.1 Test Instruction Declaration	20
6.2.2 Test Command List	20
Chapter 7 Hardware Design	20
Chapter 8 Frequently Asked Questions	21
8.1 Transmission distance is not ideal	21
8.2 Module is easily damaged	21
8.3 The bit error rate is too high.	22
8.4 Antenna Selection	22
Chapter 9 Welding Operation Instructions	22
9.1 Reflow soldering temperature	22
9.2 Reflow Soldering Profile	23
Chapter 10 Antenna Guide	23
1.0.1 Antenna Recommendation	23
Revision History	24
about Us	24

Disclaimer and Copyright Notice

The information contained herein, including URLs provided for reference, is subject to change without notice. This document is provided "as is" without warranty of any kind, including any warranty of merchantability, fitness for a particular purpose, or non-infringement, and any warranty of any proposal, specification, or sample mentioned elsewhere. This document assumes no liability whatsoever, including liability for any infringement of any patent rights arising from the use of the information contained herein. This document does not grant any license to use any intellectual property rights herein, whether express or implied, by estoppel or otherwise.

The test data obtained in this article are all from tests conducted by EBYTE Labs, and the actual results may differ slightly.

All trademark names, trademarks and registered trademarks mentioned in this article are the property of their respective owners.

The final interpretation right belongs to Chengdu Ebyte Electronic Technology Co., Ltd.

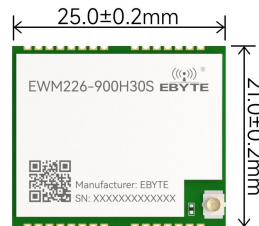
Notice:

Due to product version upgrades or other reasons, the content of this manual may change. Ebyte Electronics Technology Co., Ltd. reserves the right to modify the content of this manual without prior notice or warning. This manual is for guidance only. Ebyte Electronics Technology Co., Ltd. has made every effort to provide accurate information in this manual; however, Ebyte Electronics Technology Co., Ltd. does not guarantee that the content of this manual is completely error-free, and all statements, information, and suggestions in this manual do not constitute any express or implied warranty.

Chapter 1 Product Overview

1.1 Product Introduction

LoRa wireless frequency hopping data transmission module for lawnmowers designed based on LoRa spread spectrum technology . It has a maximum transmission power of 30 dBm, multiple transmission modes, operates in the 868/915MHz frequency band , and has TTL level output. It is suitable for applications such as lawnmowers and surveying .



1.2 Features and Functions

- It adopts a new generation of LoRa spread spectrum modulation technology, which has a longer communication distance and stronger anti-interference ability;
- multiple frequency bands including 868/915MHz (customizable) ;
- Supports fixed-frequency and frequency-hopping transmission modes;
- Supports one-to-many communication;
- Supports serial port firmware upgrades , making maintenance more convenient;
- It supports AT commands, making it convenient and quick for users to use ;
- It supports users setting their own communication keys, which cannot be read, greatly improving the confidentiality of user data;
- It supports LBT function, which listens to channel environmental noise before transmission, which can greatly improve the communication success rate of the module in harsh environments;
- Supports RSSI signal strength indication function for evaluating signal quality;
- Supports fixed-point transmission, broadcast transmission, and channel monitoring;
- The air transmission rate is adjustable in multiple levels;
- The parameters are saved after power failure, and the module will work according to the set parameters after power is restored;
- Built-in watchdog timer automatically resets and resumes operation in case of an accident;

1.3 Application Scenarios

- Lawn mower applications;
- Drone applications;
- Intelligent irrigation and environmental monitoring in agriculture;
- Photovoltaic power plant monitoring and distributed photovoltaic system management, etc.
- Smart home and industrial sensors, etc.;
- Building automation solutions;
- Wireless industrial-grade remote control .

Chapter 2 Specifications

2.1 Product Selection Comparison

Product Model	Carrier frequency ① M Hz	Transmit power ② dBm	Test distance ③ km	Packaging	Product dimensions (mm)	Antenna type ④
EWM226-900H30S	860 ~ 930	30	5	patch	21.0 * 25.0	IPEX 1st Generation / Stamp Hole

Note:

- (1)Carrier frequency ① : Frequency band range can be customized by the user;
- (2)Transmit power ② : 30dBm, error range ± 1 dBm, power is adjustable in multiple levels, please see [Chapter 6 of this article for details of power levels](#) ;
- (3)Test distance ③ : Clear and open, airspeed 11kbps, antenna height 2 meters (for reference only, actual measurement is recommended);
- (4)Antenna type ④ : Equivalent impedance approximately 50 Ω ;

2.2 Basic parameters

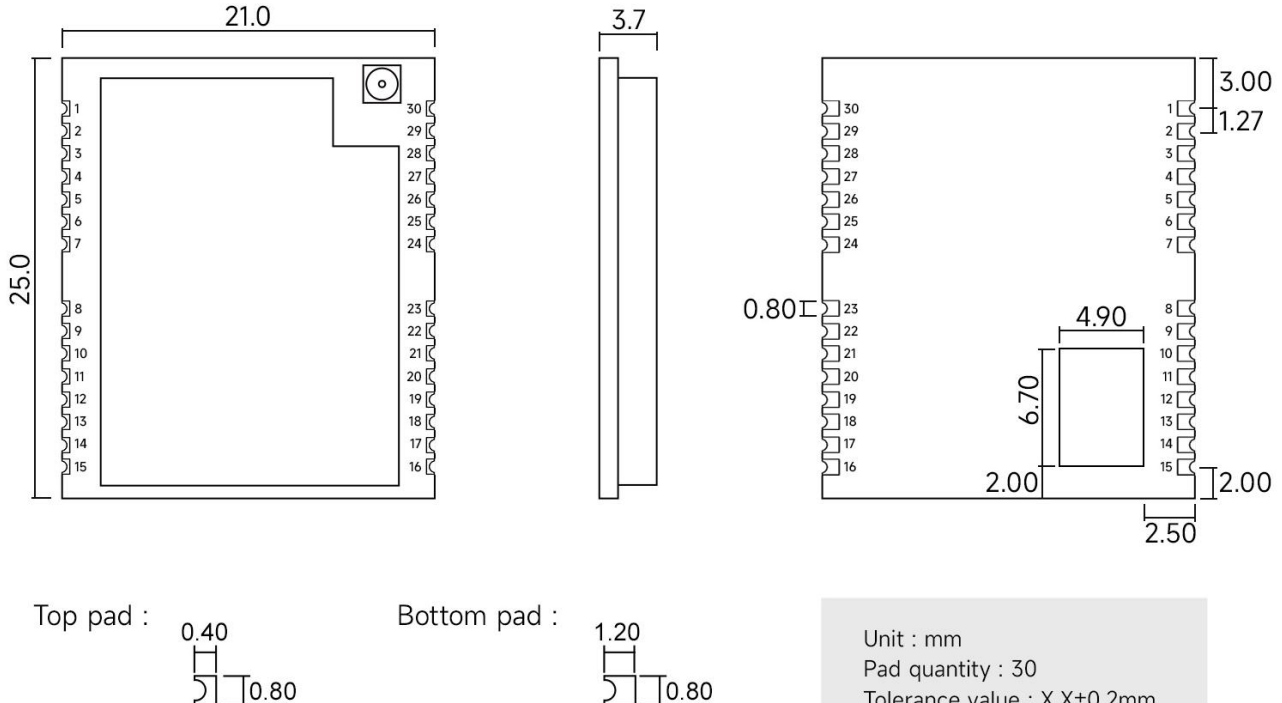
Conditions: Tc = 25°C, VCC = 3.3 V, 868MHz/915MHz

parameter	describe
Modulation method	LoRa
Interface method	1.27mm stamp perforation
Communication interface	UART
Launch length	240 bytes is the maximum capacity per packet . Packets exceeding this limit will be automatically split into smaller packets . This capacity can be adjusted via a host computer or command line. For detailed instructions, please refer to Chapter 6 of this document .
Packaging method	patch
Cache capacity	1000 Btye
weight	3.0g $\pm$ 0.1g
Work environment	
Operating temperature	-40°C $\sim$ +85°C, industrial grade standard
Operating humidity	10-90%RH
Storage temperature	-40°C $\sim$ +125°C
RF parameters	
Transmit power	30dBm = 1000mW, power adjustable in multiple levels; detailed instructions can be found in Chapter 6 of this document .
Operating	860MHz $\sim$ 930MHz, frequency band customization supported; see Chapter 6 of this document for detailed

parameter	describe
frequency band	commands.
Receiving sensitivity	- 134dBm, airspeed 2.4kbps@SF11
air speed	ranges from 11 to 62.5 Kbps, with multi-level adjustable air speeds. Detailed instructions can be found in Chapter Six of this document.
Blocking power	10dBm
Electrical parameters	
Power supply voltage	3.0~3.6V , $\geq 3.3V$ can guarantee output power, exceeding 3.6V will permanently burn out the module .
Communication level	3.0 V
Emission current	≈ 1000 mA , instantaneous power consumption, @30dBm
Received current	≈ 13 mA

Chapter 3 Mechanical Dimensions and Pin Definitions

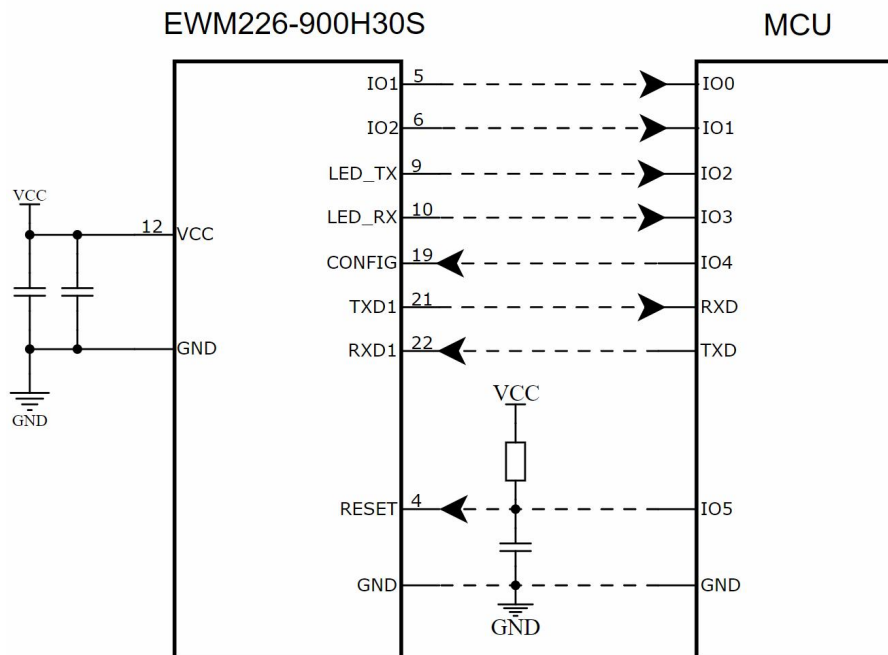
3.1 Mechanical Dimensions and Pin Definitions



Pin number	Pin Name	Pin direction	Pin Applications
1	GND	-	Module ground wire
2	TXD2	-	The feet must be suspended in the air.

Pin number	Pin Name	Pin direction	Pin Applications
3	RXD2	-	The feet must be suspended in the air.
4	RESET	enter	The module reset pin requires a low-level reset, and the pull-low time must be greater than 100μs.
5	IO1	Output	The RXD1 buffer is an empty indicator pin; a high level indicates it is empty, and a low level indicates it is not empty. The size of each serial port data packet cannot exceed 1000 bytes.
6	IO2	Output	The master-slave module synchronization indicator pin will output a 1Hz square wave signal after synchronization.
7	GND	-	Module ground wire
8	GND	-	Module ground wire
9	LED_TX	Output	The RF transmit indicator pin outputs a high level when transmitting data and a low level when idle.
10	LED_RX	Output	The RF receiver indicator pin outputs a high level when receiving data and a low level when idle.
11	GND	-	Module ground wire
12	VCC	enter	Power supply interface, DC 3.0 ~ 3.6V
13	VCC	enter	Power supply interface, DC 3.0 ~ 3.6V
14	GND	-	Module ground wire
15	GND	-	Module ground wire
16	GND	-	Module ground wire
17	SWDIO	-	The feet must be suspended in the air.
18	SWCLK	-	The feet must be suspended in the air.
19	CONFIG	enter	The mode configuration pin is low to enter configuration mode and high to enter transmission mode.
20	1PPS	enter	1PPS input pin
twenty one	TXD1	Output	Serial output pin
twenty two	RXD 1	enter	Serial input pin
twenty three	GND	-	Module ground wire
twenty four	GND	-	Module ground wire
25	NC	-	The feet must be suspended in the air.
26	NC	-	The feet must be suspended in the air.
27	RSV	-	Reserved pins must be left floating.
28	GND	-	Module ground wire
29	ANT	-	Antenna interface (50 ohms characteristic impedance)
30	GND	-	Module ground wire

Chapter 4 Recommended Connection Diagram



Serial Number	Precautions
1	The module has a 3.3V serial port and I/O pins. If the user is using a 5V microcontroller, level conversion needs to be considered.
2	It is strongly recommended to connect the RESET pin to avoid unexpected malfunctions or other special situations.
3	If you are considering adding an external switch to the module's VCC, please consider the leakage current of the external microcontroller's serial port; otherwise, the module will not be able to be turned off.
4	(Optional) For surface mount modules, consider bringing out the module's programming pins for easier future maintenance or firmware customization.
5	(Optional) Consider bringing out an additional interface for the module's serial port to facilitate future maintenance (can be upgraded online with PC tools).

Chapter 5 Functional Details

- In this chapter , the CONFIG pin is described as follows: 0 represents a low level and 1 represents a high level.

5.1 Working Mode

The module has two operating modes, which are set by the CONFIG pin . Details are shown in the table below:

Pattern (0-1)	CONFIG	Mode Introduction	Remark
0 Configuration mode	0	Module parameter configuration	-
1 Transmission Mode	1	Data transmission modes (DUPLEX, TXONLY, RXONLY, HOP)	Switch using the AT+MOD command.

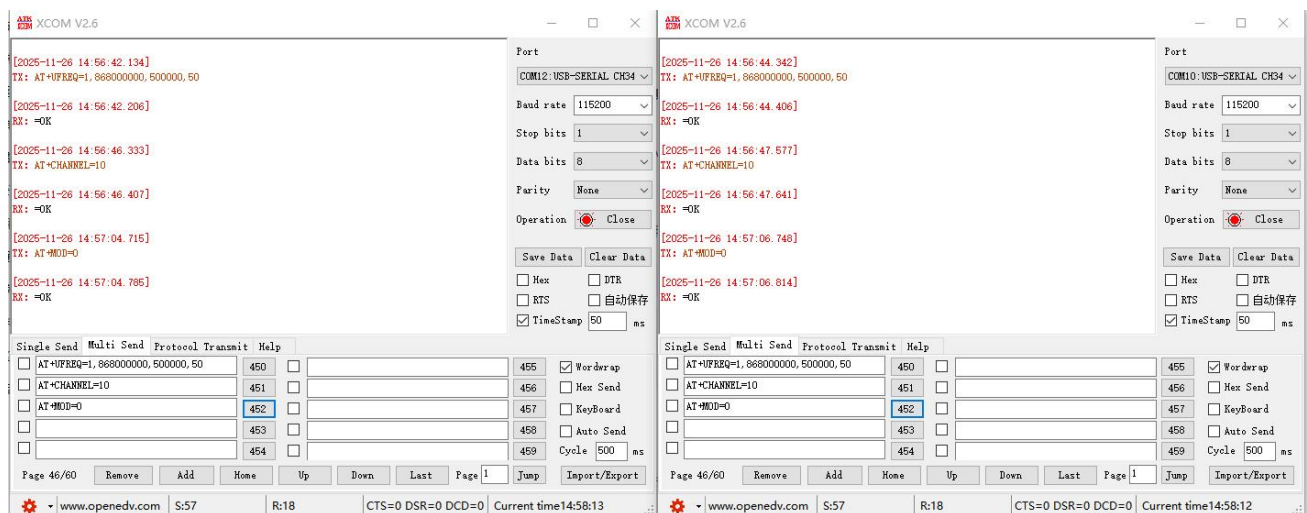
5.1.1 Transmission mode

5.1.1.1 Quick Start Guide to DUPLEX Transfer

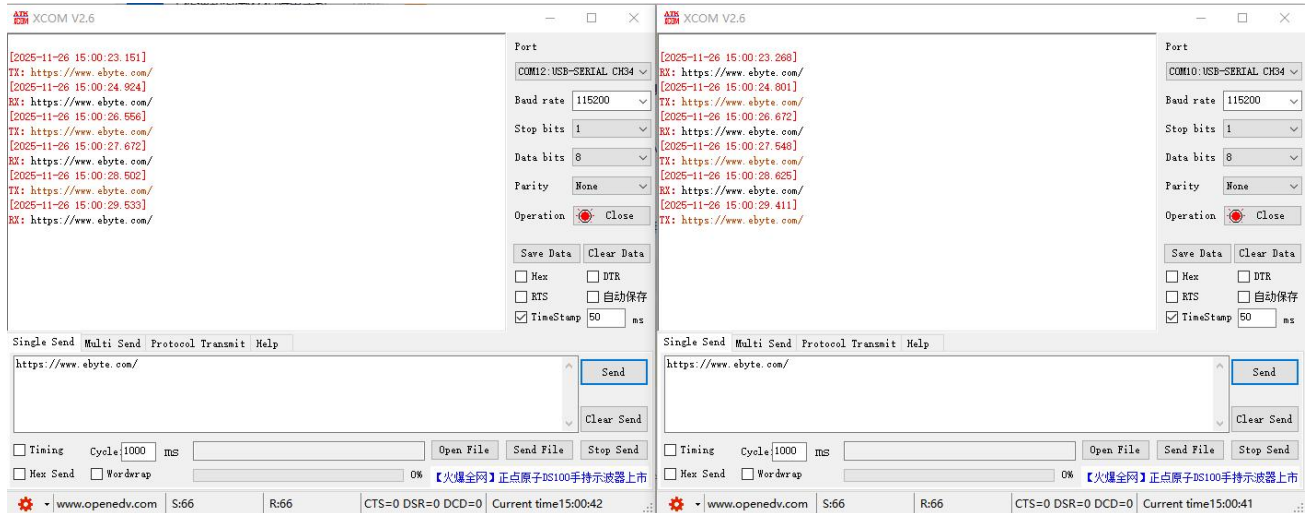
DUPLEX Mode: In this mode, bidirectional data transmission is possible (the module can both send and receive wireless data).

What you send is what you get; you can communicate with each other using a serial port assistant. An example is shown below:

- ① Switch the CONFIG pin to configuration mode;
- ② AT+UFREQ =1,868000000,500000,50 (Configure fixed-frequency transmission, custom frequency start for reception, channel spacing, and number of channels);
- ③ AT+CHANNEL=10 (Configure fixed-frequency transmit and receive frequency FREQ = 868000000 + 500000*10 Hz);
- ④ AT+MOD=0 (Set the transfer mode to DUPLEX);
- ⑤ Simply switch the CONFIG pin to transmission mode.



Configuration diagram



Communication diagram

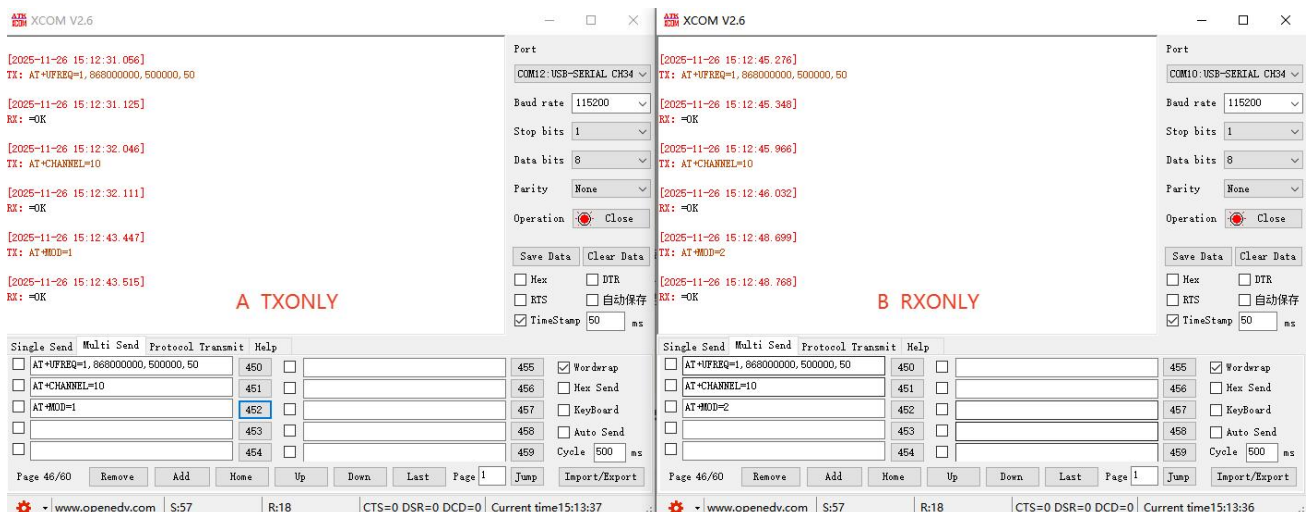
5.1.1.2 Quick Use of TXONLY and RXONLY

TXONLY mode: In this mode, only data can be transmitted (the module can only transmit wireless data and cannot receive wireless data), and it needs to be used in conjunction with RXONLY or DUPLEX mode.

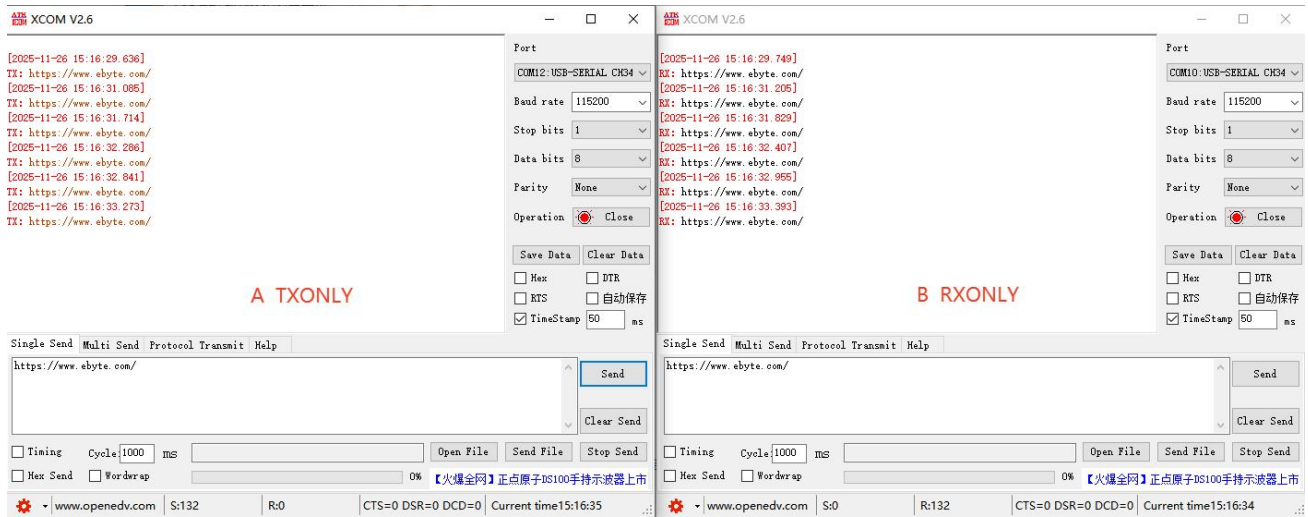
RXONLY mode: In this mode, the module can only receive data (it can only receive wireless data and cannot send wireless data). It needs to be used in conjunction with TXONLY or DUPLEX mode.

T XONLY and RXONLY :

- ① Switch the CONFIG pin to configuration mode;
- ② $AT+UFREQ=1,868000000,500000,50$ (Configure fixed-frequency transmission, custom frequency start for reception, channel spacing, and number of channels);
- ③ $AT+CHANNEL=10$ (Configure fixed-frequency transmit and receive frequency $FREQ = 868000000 + 500000 \times 10$ Hz);
- ④ **Module A** : $AT+MOD=1$ (Set transmission mode to TXONLY) **Module B** : $AT+MOD=2$ (Set transmission mode to RXONLY)
- ⑤ Simply switch the CONFIG pin to transmission mode.



Configuration diagram



Communication diagram

5.1.1.3 Quick Use of HOP Mode

Detailed explanation of frequency hopping mode parameters:

①AT+ROLE

In HOP mode, master-slave role configuration supports one master and one slave, or one master and multiple slaves.

Master (host): After the CONFIG pin is switched to transmission mode, the host begins to send frequency hopping synchronization frame data at a frequency of once per second;

From the slave role (slave): After the CONFIG pin switches to transmission mode, the slave starts fast frequency hopping to search for master synchronization data until it receives normal master synchronization frame data, and then immediately begins synchronizing the master frequency hopping sequence. After synchronization is complete, the IO2 and LED_RX pins will synchronize the master output. At this time, the master and slave have completed frequency hopping synchronization, and the user can start data pass-through communication.

②AT+FLIST

In HOP mode, the frequency hopping list can be selected based on different countries and regions. (European, American, and Australian standards are supported; the master and slave devices must be consistent).

③AT+PPS

In HOP mode, the external pulse-per-second input enable switch is off by default, and the module's internal software clock automatically generates a synchronization pulse signal per second. If the user enables this switch, external hardware must be connected to the module's 1PPS pin to provide a stable synchronization pulse per second for the module's frequency hopping system (commonly used for PPS pin access in positioning devices).

④AT+WTIME

In HOP mode, the slave data return window time configuration is disabled by default, allowing only the master to transmit data to the slave. If a user needs the slave to return data to the master in HOP mode, both the master and slave must enable this window time and ensure it remains consistent. Users can adjust the return window according to their slave needs; a larger return window results in a larger amount of data returned by the slave, while conversely, the amount of data sent by the master will decrease accordingly. Users should adjust this based on their usage scenario.

⑤AT+ADDR

In all transmission modes, the module address configuration must ensure that the module addresses of both communicating parties are consistent; otherwise, communication will not be possible.

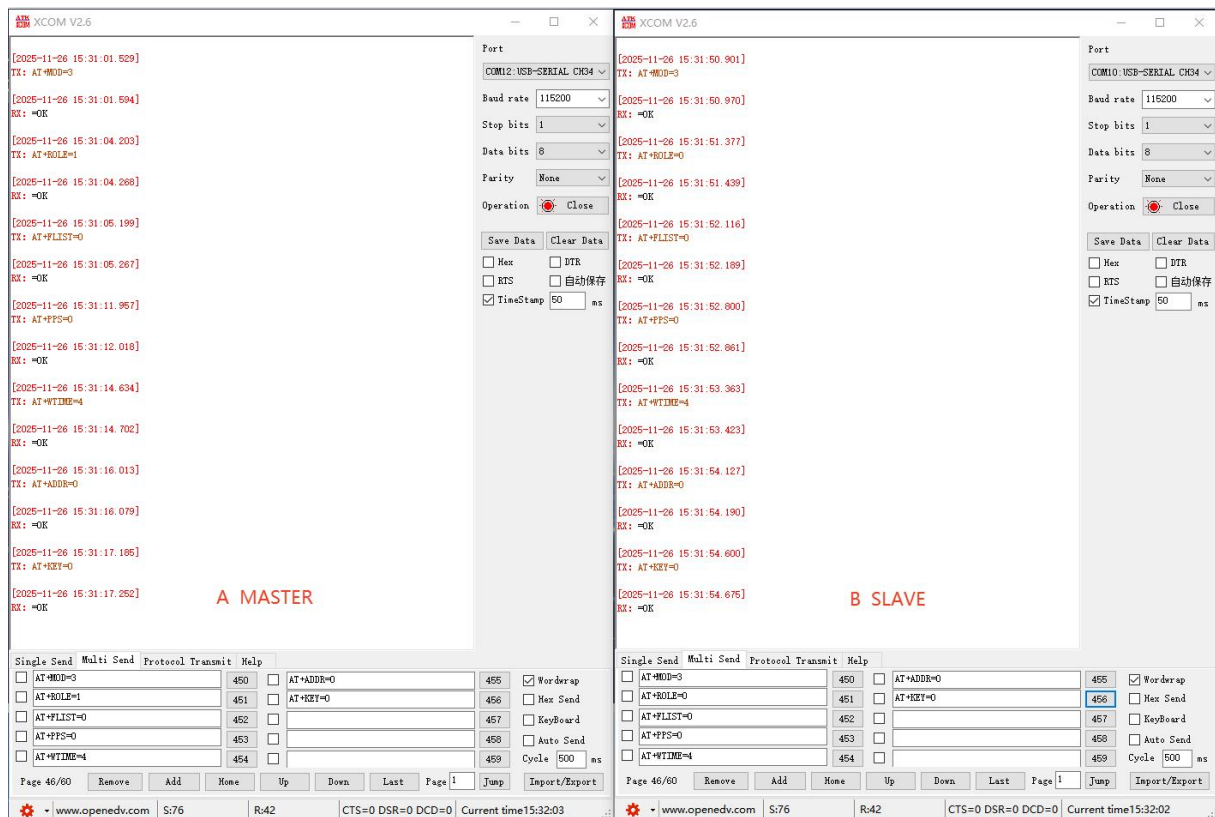
⑥AT+KEY

In all transmission modes, the module data transmission key configuration must ensure that the module keys of both communicating parties are consistent; otherwise, communication will not be possible.

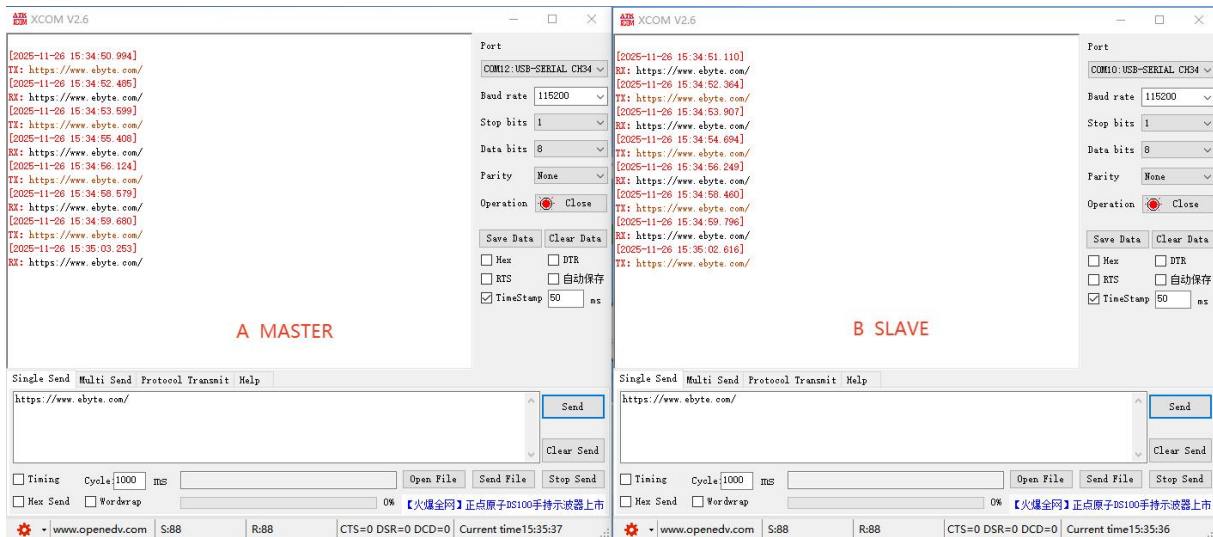
Operating steps:

- Example scenarios and steps are as follows:

- ① The CONFIG pin is switched to configuration mode;
- ② AT+MOD=3 (Set frequency hopping-HOP transmission mode);
- ③ **Module A** : AT+ROLE=1 (Set the frequency hopping transmission master role), **Module B** : AT+ROLE=0 (Set the frequency hopping transmission slave role);
- ④ AT+FLIST=0 (Set the frequency hopping list to ensure consistency between master and slave roles);
- ⑤ AT+PPS=0 (Sets whether to enable hardware synchronization; requires external hardware support to ensure master-slave consistency).
- ⑥ AT+WTIME=4 (Sets the slave data return cycle. The master and slave must maintain consistency. The longer the cycle, the more data the slave returns.)
- ⑦ AT+ADDR=0 (Custom address, master and slave can broadcast to multiple slaves with consistent operation, but cannot send back to one slave from multiple slaves).
- ⑧ AT+KEY=0 (Custom communication encryption, master and slave remain consistent, frequency hopping sequence is related to this);
- ⑨ When the CONFIG pin is switched to transmission mode, the master and slave modules will start to synchronize automatically. After synchronization is completed, the master and slave IO2 pins will output a 1Hz square wave signal, and transparent data communication can begin.



Configuration diagram



Communication diagram

5.2 Serial port firmware upgrade

- ① If customers need to upgrade the firmware, they need to contact the relevant person through the sales channel to obtain the corresponding upgrade package, and then use the host computer provided by EBITE to upgrade the firmware. Under normal circumstances, users do not need to upgrade the firmware, so **please do not use the "AT+IAP" command**.
- ② The necessary pins for the upgrade must be brought out (CONFIG, TXD, RXD, VCC, GND), and then the "AT+IAP" command should be sent in configuration mode to enter upgrade mode.
- ③ After entering upgrade mode, the baud rate will automatically switch to 115200 until it automatically exits, during which time logs will be output.

5.2.1 Upgrade your host computer quickly

Step 1: Enter configuration mode by changing CONFIG (Note: the baud rate in configuration mode is 115200); open the configuration host computer and select the firmware upgrade tab.

Step 2: Click "Read Parameters" to view module information in the left window of the host computer.

Step 3: Open the file (select the firmware.bin file) > Click Start Download.

Step 4: Click "Start Download". The firmware upgrade will begin and a message will appear indicating successful download upon completion.

Step 5: After the firmware upgrade is complete, reopen the serial port to confirm the read parameters.



upper computer interface diagram

Chapter 6 AT Commands

- Parameter configuration or querying using AT commands must be done in configuration mode;
- Automatically recognizes the carriage return and newline character at the end of user AT commands, and still recognizes them even if no carriage return and newline character is added;
- An AT command recognition or execution error will result in an "**=ERR**" response.
- When the input parameters exceed the range, they will be restricted. Please do not allow the parameters to exceed the range to avoid unforeseen situations.

6.1 General instructions

6.1.1 AT+RESET soft reboot

Instruction format	AT+RESET
Execution Example	Send: AT+RESET Response: =OK
Feature Description	● The actual reset process will only begin after the user responds (=OK). Please wait approximately 30ms before proceeding to the next step.

6.1.2 AT+DEFAULT to restore factory settings

Instruction format	AT+DEFAULT
Execution Example	Send: AT+DEFAULT Response: =OK
Feature Description	● After the response (=OK), in addition to parameter restoration, a restart process will be executed. Please wait approximately 30ms before proceeding to the next step.

6.1.3 Use AT+DEVTYPE to query product model

Instruction format	AT+DEVTYPE=?
Execution Example	Send: AT+DEVTYPE=? Response: DEVTYPE=E WM226-900H30S
Feature Description	● Product Series: EWM226 ● Operating frequency band: 900 ● Maximum power: 30

6.1.4 AT+FWCODE Query Software Version

Instruction format	AT+FWCODE=?
Execution Example	Send: AT+FWCODE=? Response: FWCODE= 7547-0-10
Feature Description	● Major version number: 7547 ● Branch number: 0 ● Minor version number: 10 (incrementing from 10)

6.1.5 AT+ IAP Serial port firmware upgrade

Instruction format	AT+IAP
Setup Example	Send: AT+IAP Response: =OK
Query Example	Send: AT+IAP Response: =OK
Parameter Description	none
Feature Description	● Use with caution

6.1.6 AT+MOD Transmission Working Mode

Instruction format	AT+MOD=<mode>
Setup Example	Send: AT+MOD=0 Response: =OK
Query Example	Send: AT+MOD=? Response: AT+MOD=0
Parameter Description	<mode> : Working mode number 0 DUPLEX (Fixed Frequency Transceiver Mode) 1 TXONLY (Fixed Frequency Single Transmission Mode) 2 RXONLY (Fixed Frequency Single Receiver Mode) 3 HOP (Frequency Hopping Mode)
Feature Description	● DUPLEX, TXONLY, and RXONLY are fixed-frequency transmission modes. ● HOP is a frequency hopping transmission mode.

6.1.7 AT+ADDR Communication Address (Match Filtering)

Instruction format	AT+ADDR=<address>
Setup Example	Send: AT+ADDR=0 Response: =OK
Query Example	Send: AT+ADDR=? Response: AT+ADDR=0
Parameter Description	<address> : Communication address 0 - 65535
Feature Description	● Default address 0 ● The address is not a radio frequency characteristic; it is merely an identifier carried within the data packet to help the receiving software logic determine whether to perform match filtering. ● The broadcast address is 65535. As a receiver, it can receive data from any address on the same channel (without filtering). As a sender, the data packets it sends carry the broadcast address, so the recipient will not filter them (broadcast data).

6.1.8 AT+NETID Communication Network Code (Matching Filtering)

Instruction format	AT+NETID=<net>
Setup Example	Send: AT+NETID=0 Response: =OK
Query Example	Send: AT+NETID=? Response: AT+NETID=0
Parameter Description	<net> : Communication network code 0-255
Feature Description	● Default network 0 ● Network codes are not radio frequency characteristics; they are merely identifiers carried within data packets. ● Network codes have lower filtering priority than broadcast addresses. Even if the network codes are different, broadcast data can still be received. ● This instruction is reserved; customers do not need to use it at this time.

6.1.9 AT+CHANNEL Communication Channel (Carrier Frequency)

Instruction format	AT+CHANNEL=<ch>
Setup Example	Send: AT+CHANNEL=0 Response: =OK
Query Example	Send: AT+CHANNEL=? Response: AT+CHANNEL=0
Parameter Description	Based on AT+UFREQ =p1,p2,p3,p4 configuration parameters
Feature Description	actual carrier frequency = p2 + ch * p3 (Hz)

6.1.10 AT+KEY Communication Password (Data Encryption)

Instruction format	AT+KEY=<key>
Setup Example	Send: AT+KEY=0 Response: =OK
Parameter Description	<key> : Encryption/decryption password 0 - 65535
Feature Description	- Default 0 - Unable to query (password cannot be read) - Modules with inconsistent passwords cannot communicate (they cannot decode the correct data packets, fail verification, and are discarded).

6.1.1 AT +UART Serial Port Baud Rate and Verification

Instruction format	AT+UART=<baud>,<parity>
Setup Example	Send: AT+UART=3,0 Response: =OK
Query Example	Send: AT+UART=? Response: AT+UART=3,0
Parameter Description	<baud> : Serial port baud rate 0 1200bps 1 2400bps 2 4800bps 3 9600bps 4 19200bps 5 38400bps 6 57600bps 7 115200bps (default) <parity> : Serial port verification type 0 No validation None (default) 1. Odd parity 2 Even parity
Feature Description	The change will take effect after switching working modes.

6.1.1 2 AT+PACKET Serial Port Packing Length

Instruction format	AT+PACKET=<len>
Setup Example	Send: AT+PACKET=0 Response: =OK
Query Example	Send: AT+PACKET=? Response: AT+PACKET=0
Parameter Description	<len> : Serial port data packet length 0 240 bytes (default) 1 128 bytes 2 64 bytes 3 32 bytes

Feature Description	● If the user-input data is less than the set length during serial port reception, it will be handled as a timeout frame break, then packaged and transmitted wirelessly. ● During serial port reception, when the user-input data reaches the set length, it will be immediately packaged and wirelessly transmitted. ● Data exceeding the set length will continue to accumulate in the serial port buffer. After wireless transmission is complete, it will be split into packets according to the set length and transmitted in a cyclical manner. ● In HOP mode, it is recommended to keep the default values.
---------------------	---

6.1.1 3 AT +URXT Serial Port Frame Drop Conditions

Instruction format	AT+URXT=<byte>
Setup Example	Send: AT+URXT=3 Response: =OK
Query Example	Send: AT+URXT=? Response: AT+URXT=3
Parameter Description	<byte> : byte time 1 - 255
Feature Description	● By default, serial port timeout frame interruption is handled based on 3-byte intervals. ● The time (per byte) varies at different baud rates. At a baud rate of 9600, it's approximately 1 millisecond per byte; at a baud rate of 115200, it's approximately 100 microseconds per byte. ● When some customers input data from their external microcontrollers via serial ports to the module, the data is discontinuous (e.g., due to RTOS task interruption). This exceeds the module's set byte time window, resulting in frame fragmentation and manifesting as segmented data packets at the receiving end.

6.1.1 4 AT +RATE Wireless Transmit/Receive Rate

Instruction format	AT+RATE=<rate>
Setup Example	Send: AT+RATE=2 Response: =OK
Query Example	Send: AT+UART=? Response: AT+UART=2
Parameter Description	<rate> : air speed 0 11Kbps (default) 1 19.2Kbps 2 62.5Kbps
Feature Description	● The airspeed range varies depending on the frequency band; please pay attention to the differences. ● The higher the air speed, the lower the communication latency (shorter transmission time), but the worse the receiving sensitivity (shorter communication distance).

6.1.1 5 AT+UFREQ Setting the operating frequency parameters

Instruction format	AT+UFREQ=<mode>
Setup Example	Send: AT+UFREQ =1,868000000,500000,50 Response: =OK
Query Example	Send: AT+ UFREQ =? Response: AT+ UFREQ = 1,868000000,500000,50
Parameter Description	AT+UFREQ =p1,p2,p3,p4 <p1> Enable custom switch: 0: Off; 1: On <p2> 868000000 Sets the module's starting operating frequency to 868MHz <p3> 500000 Channel spacing 500kHz <p4> 50 Maximum number of channels: 50
Feature Description	● After setting this parameter, it can be used in conjunction with the AT+CHANNEL command .

6.1.1 6 AT+LBT Listen before you speak (avoidance)

Instruction format	AT+LBT=<on_off>
Setup Example	Send: AT+LBT=0 Response: =OK
Query Example	Send: AT+LBT=? Response: AT+LBT=0
Parameter Description	<on_off> : Switch 0. Off (default) 1. Turn on
Feature Description	- When the LBT (Listen Before Talk) function is enabled, it will check whether the ambient signal strength is lower than a set threshold before each wireless data transmission. If the condition is met within the timeout period, wireless transmission will actually start. If the condition is still not met after the maximum timeout period, wireless transmission will be forcibly started. - For specific LBT settings, please refer to the AT+LBR command.

6.1.1 7 AT+LBR Listen-before-send Judgment Condition

Instruction format	AT+LBR=<rss>,<timeout>
Setup Example	Send: AT+LBR=-55,2000 Response: =OK
Query Example	Send: AT+LBR=? Response: AT+LBR=-55,2000
Parameter Description	<rss> : Signal strength threshold (dBm) 0 - (-128) <timeout> : Maximum timeout (milliseconds) 0 - 65535
Feature Description	The default parameters are -55dBm and 2000ms.

6.1.18 AT+ROLE HOP characters

Instruction format	AT+ROLE=<role>
Setup Example	Send: AT+ROLE=0 Response: =OK
Query Example	Send: AT+ROLE=? Response: AT+ROLE=0
Parameter Description	<role> : character 0 Slave (default) 1. Host
Feature Description	Master-slave configuration in HOP mode

6.1.19 AT+WTIME Slave Data Transmission Cycle

Instruction format	AT+WTIME=<timeout>
Setup Example	Send: AT+WTIME=3 Response: =OK
Query Example	Send: AT+WTIME=? Response: AT+WTIME=3
Parameter Description	<timeout> : Sleep time 0. Off (default) 1 60ms 2 160ms 3 260ms 4 360ms
Feature Description	Both the master and the servant need to maintain consistency.

6.1.20 AT+ERSSI Ambient Signal Strength

Instruction format	AT+ERSSI=<on_off>
Setup Example	Send: AT+ERSSI=0 Response: =OK
Query Example	Send: AT+ERSSI=? Response: AT+ERSSI=0
Parameter Description	<on_off> : Switch 0. Off (default) 1. Turn on
Feature Description	RSSI Ambient Noise Enable " section in the register description description chapter and use hexadecimal instructions to read the data.

6.1.2 1 AT+ DRSSI Received Data Packet Signal Strength

Instruction format	AT+DRSSI=<on_off>
Setup Example	Send: AT+DRSSI=0 Response: =OK
Query Example	Send: AT+ERSSI=? Response: AT+ERSSI=0
Parameter Description	<on_off> : Switch 0. Off (default) 1. Turn on
Feature Description	- When enabled, signal strength data (1 byte, hexadecimal, signed number) will be appended to the end of the received data packet. - Conversion Example: Assuming the obtained additional signal strength data is 0xC4 (-60dBm), the hexadecimal value 0xC4 is 196 in decimal. The conversion formula is: RSSI = -(256 - 196) = -60 - If the distance between the sender and receiver is too short or the sender's power is too high, the received signal strength may be 0 (upper limit), please directly convert it to 0dBm.

6.1.22 AT+HPASW High Power Switch

Instruction format	AT+HPASW=<on_off>
Setup Example	Send: AT+HPASW=0 Response: =OK
Query Example	Send: AT+HPASW=? Response: AT+HPASW=0
Parameter Description	<on_off>: Switch 0. Off (default) 1. Turn on
Feature Description	Once activated, the module's maximum power can reach 30dBm.

6.1. 2 3 AT+POWER power rating

Instruction format	AT+POWER = <Power>
Setup Example	Send: AT+POWER=0 Response: AT+POWER=0

Query Example	Send: AT+POWER=? Response: AT+POWER=0	
Parameter Description	<Power> : Power AT+HPASW=1 0 30dBm (default) 1 27dBm 2 24dBm 3 21dBm	<Power> : Power AT+HPASW=0 0 22dBm (default) 1 17dBm 2 14dBm 3 10dBm
Feature Description	-	

6.1.24 AT+FLIST HOP Mode Frequency List

Instruction format	AT+FLIST =<on_off>
Setup Example	Send: AT+FLIST=0 Response: =OK
Query Example	Send: AT+FLIST=? Response: AT+FLIST=0
Parameter Description	<on_off> : Switch 0. European Standard (Default) 1 US Standard 2. Australian regulations
Feature Description	- European standard: 863MHz-870MHz 863200000, 863550000, 863900000, 864250000, 864600000, 864950000, 865300000, 865650000, 866000000, 866350000, 866700000, 867050000, 867400000, 867750000, 868100000, 868450000, 868800000, 869150000, 869500000, 869850000, - US standard: 902MHz-928MHz 902500000, 903010000, 903520000, 904030000, 904540000, 905050000, 905560000, 906070000, 906580000, 907090000, 907600000, 908110000, 908620000, 909130000, 909640000, 910150000, 910660000, 911170000, 911680000, 912190000, 912700000, 913210000, 913720000, 914230000, 914740000, 915250000, 915760000, 916270000, 916780000, 917290000, 917800000, 918310000, 919330000, 920350000, 920860000, 918820000, 919840000, 921370000, 921880000, 922390000, 922900000, 923410000, 923920000, 924430000, 924940000, 925450000, 925960000, 926470000, 926980000, 927490000, - Australian standard: 915MHz-928MHz 915500000, 916200000, 916900000, 917600000, 918300000, 919000000, 919700000, 920400000, 921100000, 921800000, 922500000, 923200000, 923900000, 924600000, 925300000, 926000000, 926700000, 927400000,

6.1.25 AT+SYNCT HOP slave synchronization timeout

Instruction format	AT+SYNCT=<time>
Setup Example	Send: AT+SYNCT=5 Response: =OK
Query Example	Send: AT+SYNCT=? Response: AT + SYNCT = 5
Parameter Description	<time> : Timeout period 2 - 64 seconds
Feature Description	- Default synchronization timeout is 5 seconds - In HOP mode, after the master and slave devices complete synchronization, if the slave device does not receive synchronization frame data from the master device within the set timeout period, the slave device will

automatically clear the synchronization state and re-enter the fast frequency hopping reception mode to begin resynchronization.

- **Users are advised to use this with caution (keep the default settings).**

6.2 Signaling testing (certification)

6.2.1 Test Instruction Declaration

The AT command testing section is intended for use only as an aid in product certification (e.g., CE, FCC certification) and should not be used for other purposes.

Ebyte assumes no liability for any damages arising from failure to comply with relevant operating or design specifications. Ebyte endeavors to ensure the completeness, accuracy, and timeliness of features under development, but does not exclude the possibility of errors or omissions in such features. Unless otherwise agreed, no implied or statutory warranties are made regarding the use of features under development. To the maximum extent permitted by applicable law, Ebyte shall not be liable for any damages suffered as a result of using features under development, whether or not such damages were foreseeable.

6.2.2 Test Command List

- Detailed instructions and documentation can be obtained through Ebitech's sales channels.

AT commands	Function Summary
AT+FPOWER	Fine-tuning of transmit power
AT+XCAP	Crystal oscillator calibration
AT+ULORAM	LoRa modulation parameters modification (SF spreading factor, BW transmit bandwidth, CR coding rate)
AT+OPTIMIZE	Low-rate optimization options
AT+SWAVE	Fixed-frequency continuous single-carrier transmission
AT+MWAVE	Fixed-frequency continuous modulation wave transmission
AT+TFHSS	Frequency hopping modulated wave transmission
AT+DUTYTX	Fixed-frequency continuous single-carrier/modulated wave duty cycle transmission

Chapter 7 Hardware Design

- It is recommended to use a DC regulated power supply to power this module, with the power supply ripple coefficient as small as possible, and the module must be reliably grounded;
- Please ensure the correct connection of the positive and negative power terminals. Reversing the connection may cause permanent damage to the module.
- Please check the power supply and ensure it is within the recommended voltage range. Exceeding the maximum value will cause permanent damage to the module.
- Please check the power supply stability; the voltage should not fluctuate significantly or frequently.
- When designing power supply circuits for modules, it is often recommended to retain a margin of more than 30% to ensure long-term stable operation of the entire unit.
- The module should be kept as far away as possible from power supplies, transformers, high-frequency traces, and other parts with high electromagnetic interference.
- High-frequency digital traces, high-frequency analog traces, and power traces must avoid the area under the module. If it is absolutely necessary to run under the module, assuming the module is soldered on the Top Layer, lay ground copper on the Top Layer of the module contact area (all copper and well grounded), and run the traces close to the digital part of the module on the

Bottom Layer.

- Assuming the module is soldered or placed in the Top Layer, it is also wrong to arbitrarily route traces in the Bottom Layer or other layers, as this will affect the module's stray emissions and receiver sensitivity to varying degrees.
- If there are devices around the module that cause significant electromagnetic interference, it will greatly affect the module's performance. Depending on the intensity of the interference, it is recommended to keep them away from the module. If possible, appropriate isolation and shielding can be implemented.
- If there are traces around the module that cause significant electromagnetic interference (high-frequency digital, high-frequency analog, power supply traces), it will greatly affect the module's performance. Depending on the intensity of the interference, it is recommended to keep them away from the module. If possible, appropriate isolation and shielding can be implemented.
- The antenna mounting structure has a significant impact on module performance; it is essential to ensure that the antenna is exposed and preferably vertically upward.
- When the module is installed inside the housing, a high-quality antenna extension cable can be used to extend the antenna to the outside of the housing.
- Antennas should never be installed inside a metal casing, as this will greatly reduce the transmission distance.

Chapter 8 Frequently Asked Questions

8.1 Transmission distance is not ideal

- When there are obstacles in line-of-sight communication, the communication distance will be reduced accordingly;
- Temperature, humidity, and co-channel interference can all lead to increased packet loss rates in communications.
- The ground absorbs and reflects radio waves, and the test results are poor when the ground is close to the ground.
- Seawater has a strong ability to absorb radio waves, so testing results are poor at the seaside.
- If there are metal objects near the antenna, or if it is placed inside a metal casing, the signal attenuation will be very severe.
- Power register settings are incorrect; air speed is set too high (the higher the air speed, the shorter the distance).
- At room temperature, the power supply voltage is lower than the recommended value; the lower the voltage, the lower the power output.
- The antenna may be poorly matched with the module or have quality issues.

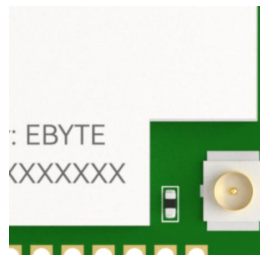
8.2 Module is easily damaged

- Please check the power supply and ensure it is within the recommended voltage range. Exceeding the maximum value will cause permanent damage to the module.
- Please check the power supply stability; the voltage should not fluctuate significantly or frequently.
- Please ensure anti-static operation during installation and use, as high-frequency devices are sensitive to electrostatic discharge.
- Please ensure that the humidity is not too high during installation and use, as some components are humidity-sensitive.
- Unless there are special requirements, it is not recommended to use it at excessively high or low temperatures.

8.3 The bit error rate is too high.

- If there is interference from a co-channel signal nearby, move away from the source of interference or change the frequency or channel to avoid the interference.
- An inadequate power supply can also cause garbled characters, so it is essential to ensure the reliability of the power supply.
- Poor quality or excessive length of extension lines or feeders can also lead to a high bit error rate.

8.4 Antenna Selection



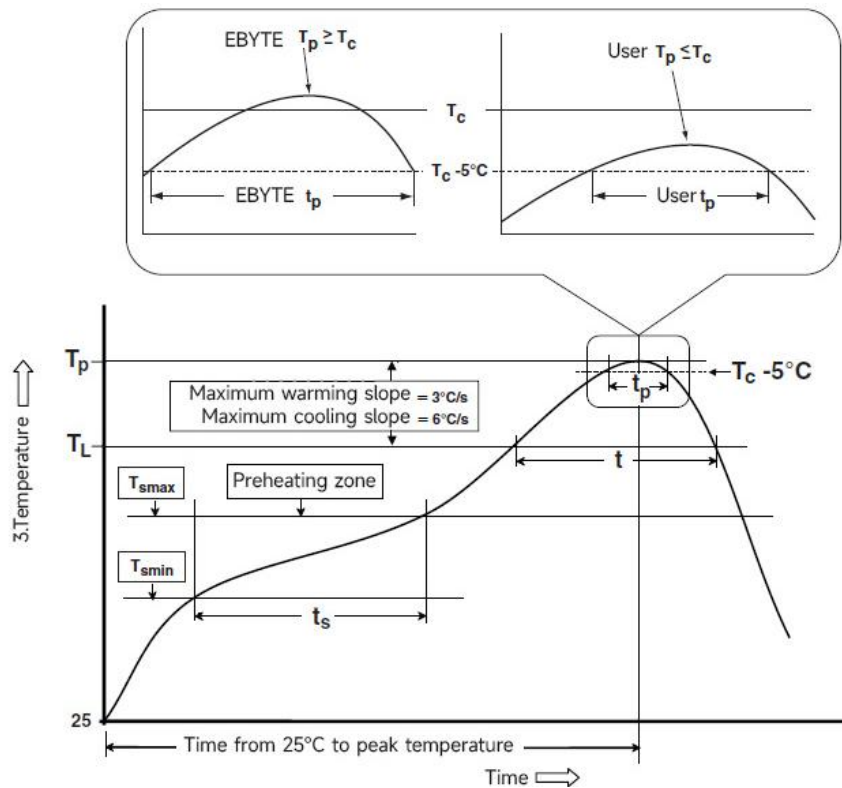
The IPEX interface and the stamp hole interface can be selected arbitrarily.

Chapter 9 Welding Operation Instructions

9.1 Reflow soldering temperature

Reflow soldering profile characteristics		Lead-based assembly	Lead-free assembly
Preheating/Insulation	Minimum temperature (T _{min})	100°C	150°C
	Maximum temperature (T _{max})	150°C	200°C
	Time (T _{min} ~T _{min})	60-120 seconds	60-120 seconds
Temperature rise slope (TL~Tp)		3°C/second, maximum value	3°C/second, maximum value
Liquid phase temperature (TL)		183°C	217°C
Duration above TL		60~ 90 seconds	60~ 90 seconds
Package peak temperature Tp		Users must not exceed the temperature specified on the product's "humidity sensitivity" label.	Users must not exceed the temperature specified on the product's "humidity sensitivity" label.
The time (Tp) within 5°C of the specified grading temperature (Tc) is shown in the figure below.		20 seconds	30 seconds
Cooling slope (Tp~TL)		6°C/second, maximum value	6°C/second, maximum value
Time from room temperature to peak temperature		6 minutes, the longest	8 minutes, the longest
* The peak temperature (Tp) tolerance of the temperature profile is defined as the user's upper limit.			

9.2 Reflow Soldering Profile



Chapter 10 Antenna Guide

1.0.1 Antenna Recommendation

Antennas play a crucial role in communication, and inferior antennas can often have a significant impact on communication systems. Therefore, our company recommends certain antennas that are high-performing and reasonably priced and compatible with our wireless modules.

Product Model	type	Frequency band (Hz)	interface	Gain (dBi)	Height (mm)	Feeder (cm)	Features
TX868-JKS-IPX20	Glue rod antenna	868M	IPEX-1	3.0	197	200	Bendable rubber rod, omnidirectional antenna
TX868-JZLW-15	Glue rod antenna	868M	IPEX-1	3.0	165	150	Bendable rubber rod, omnidirectional antenna
TX868-XPL-100	Suction Cup Antenna	868M	SMA-J	3.5	290	100	Miniature suction cup antenna, cost-effective
TX868-JKD-20	Glue rod antenna	868M	SMA-J	3.0	170	-	Bendable rubber rod, omnidirectional antenna
TX915-JKS-IPX20	Glue rod antenna	915M	IPEX-1	3.0	197	200	Bendable rubber rod, omnidirectional antenna
TX915-JZLW-15	Glue rod antenna	915M	IPEX-1	3.0	155	150	Bendable rubber rod, omnidirectional antenna

Product Model	type	Frequency band (Hz)	interface	Gain (dBi)	Height (mm)	Feeder (cm)	Features
TX915-JKD-20	Glue rod antenna	915M	SMA-J	3.5	200	-	Bendable rubber rod, omnidirectional antenna
TX915-XPL-100	Suction Cup Antenna	915M	SMA-J	3.5	260	100	Miniature suction cup antenna, cost-effective

Revision History

Version	Revision Date	Revision Notes	maintainer
1.0	2025-11-29	Initial version	Bin

about Us

Hotline: 4000-330-990

Technical support: service@cdebyte.com

Documents and RF Setting download link: <https://www.cdebyte.com>

Thank you for using Ebyte products! Please contact us with any questions or suggestions: info@cdebyte.com

Address: B5 Mould Park, 199# Xiqu Ave, High-tech District, Sichuan, China